C

LINEAR ALGEBRA • UNIT 7.01

Linear Transformations & Change of Basis

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Linear Transformations & Change of Basis Practice Exercises

1 MCQ

Let $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{bmatrix} -2 & -2 \\ -1 & 0 \end{bmatrix}$. Find $T(\langle 3, -3 \rangle)$.

A $\langle 0, -3 \rangle$

B $\langle -3, -6 \rangle$

C $\langle 0, 3 \rangle$

D $\langle 1, -2 \rangle$

2 MCQ

Which matrix represents the reflection across the line y = x?

A $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

B $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

C $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

D $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

3 MCQ

Which map is linear?

A $T(x, y) = (x + 1,\ y)$

B $T(x, y) = (xy,\ x)$

C $T(x, y) = (\sin x,\ y)$

D $T(A) = A^T$ on $M_{2\times 2}$

4 MCQ

Let $T(\mathbf{x}) = A\mathbf{x}$ with $A = \begin{bmatrix} -3 & -1 \\ 2 & -2 \end{bmatrix}$. Find $T(\langle -3, 2 \rangle)$.

A $\langle 13, -1 \rangle$

B $\langle -7, 10 \rangle$

C $\langle 8, -9 \rangle$

D $\langle 7, -10 \rangle$

5 MCQ

A 3×3 symmetric matrix T_a has eigenvalues $\lambda_1 = 20$, $\lambda_2 = 22$, and $\lambda_3 = 20$. Find the exact value of the trace $\text{tr}(T_a)$.

A 62

B 40

C 82

D 65

6 MCQ

A linear map $f : \mathrm{R}^{20} \to \mathrm{R}^{20}$ has rank $\mathrm{rank}(f) = 18$. Determine the exact value of the nullity $\dim(\ker f)$.

A 20

B 5

C 0

D 2

7 MCQ

Which matrix represents the orthogonal projection onto the x-axis?

A $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

B $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$

C $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

D $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

8 MCQ

Which map is NOT linear?

A $T(x, y, z) = (x + z,\ 0)$

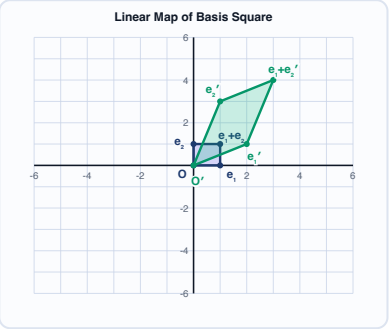
B $T(A) = \det A$ on $M_{2\times 2}$

C $T(p) = p(1)$ on P_3

D $T(A) = A^T$ on $M_{2\times 2}$

9 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.



A $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$

B $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

C $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$

D $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$

10 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation represented by the matrix $A = (2 \quad k\ 1 \quad 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

A 4

B 2

C 1

D 3

11 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation represented by the matrix $A = (3 \quad -1\ 1 \quad 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

A 5

B 6

C 4

D 7

12 MCQ

A linear map $B : \mathrm{R}^{24} \to \mathrm{R}^{24}$ has rank $\mathrm{rank}(B) = 15$. Determine the nullity $\dim(\ker B)$.

A 9

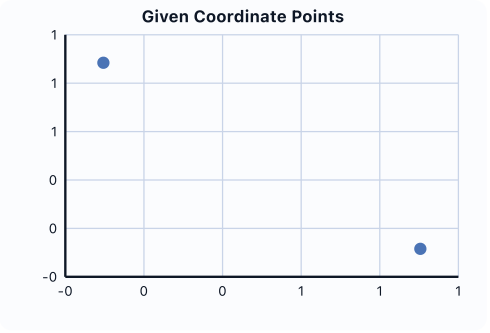
B 0

C 12

D 24

13 MCQ

Let $T : \mathrm{R}^2 \to \mathrm{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for R^2, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathrm{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for R^2, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis B is $[T]_{\mathrm{B}} = (a \quad b\ c \quad d)$, and $a + d = 5$, determine the value of $\det(T)$.



A 5

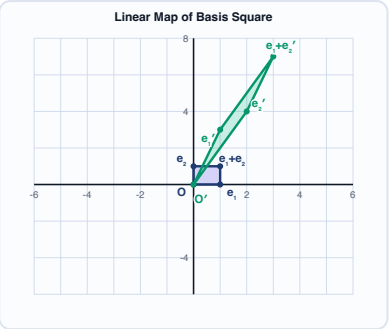
B 7

C 3

D 2

14 MCQ

Consider the linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ represented by the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. If a vector $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?



A $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$

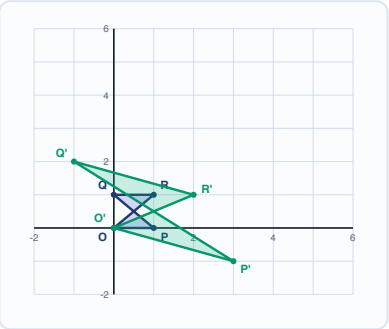
B $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

C $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$

D $\begin{pmatrix} 4 \\ 10 \end{pmatrix}$

15 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a \quad b\ c \quad d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.



A 10

B 5

C 15

D 20

16 MCQ

Let B be the basis $\langle -1, -2, 1 \rangle$, $\langle 1, -1, 0 \rangle$, $\langle 2, -1, 0 \rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v} = \langle 1, -1, 0 \rangle$.

A $\langle 0, 1, 0 \rangle$

B $\langle 1, -1, 0 \rangle$

C $\langle -2, -1, -1 \rangle$

D $\langle 0, -1, 0 \rangle$

17 MCQ

Let B be the basis $\langle 2,1,0\rangle$, $\langle 1,1,-1\rangle$, $\langle -2,-1,1\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle 4,3,-3\rangle$.

A $\langle 0,-2,1\rangle$

B $\langle 4,3,-3\rangle$

C $\langle 2,3,-3\rangle$

D $\langle 0,2,-1\rangle$

18 MCQ

Let B be the basis $\langle -1,-1,1\rangle$, $\langle -1,-2,2\rangle$, $\langle -2,1,-2\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle 8,3,-1\rangle$.

A $\langle 8,3,-1\rangle$

B $\langle -3,-1,-2\rangle$

C $\langle 2,1,9\rangle$

D $\langle 3,1,2\rangle$

19 MCQ

Let B be the basis $\langle -1,0,-1\rangle$, $\langle -2,1,2\rangle$, $\langle -2,0,-1\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle 7,0,4\rangle$.

A $\langle 4,-4,5\rangle$

B $\langle 1,0,3\rangle$

C $\langle -1,0,-3\rangle$

D $\langle 7,0,4\rangle$

20 MCQ

Let B be the basis $\langle -1,2,-1\rangle$, $\langle -2,-1,2\rangle$, $\langle 1,-1,0\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle -2,4,-2\rangle$.

A $\langle -2,0,0\rangle$

B $\langle 2,0,0\rangle$

C $\langle -2,4,-2\rangle$

D $\langle -2,-4,2\rangle$

21 MCQ

Let B be the basis $\langle 0,-1,1\rangle$, $\langle -1,-1,0\rangle$, $\langle 1,0,2\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle -2,-5,3\rangle$.

A $\langle -2,-5,3\rangle$

B $\langle -3,-2,0\rangle$

C $\langle 3,2,0\rangle$

D $\langle 4,2,0\rangle$

22 MCQ

Let B be the basis $\langle 0,-1,1\rangle$, $\langle -1,-1,1\rangle$, $\langle -2,2,-1\rangle$ of $\mathbb{R}^3$. Find $[\mathbf{v}]_B$ for $\mathbf{v} = \langle -1,4,-3\rangle$.

A $\langle -1,-1,1\rangle$

B $\langle -1,4,-3\rangle$

C $\langle 2,3,-1\rangle$

D $\langle 1,1,-1\rangle$

23 MCQ

Let B be the basis $\langle 1,-1,-1\rangle$, $\langle 0,2,-1\rangle$, $\langle -1,0,1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 6,-5,-5\rangle$.

A $\langle 6,-5,-5\rangle$

B $\langle 7,1,-6\rangle$

C $\langle -3,1,3\rangle$

D $\langle 3,-1,-3\rangle$

24 MCQ

Let B be the basis $\langle -1,1,1\rangle$, $\langle 1,0,0\rangle$, $\langle -1,2,1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 0,-6,-3\rangle$.

A $\langle 0,-6,-3\rangle$

B $\langle -6,0,-9\rangle$

C $\langle 0,3,3\rangle$

D $\langle 0,-3,-3\rangle$

25 MCQ

Let B be the basis $\langle -2,1,2\rangle$, $\langle -1,2,0\rangle$, $\langle 0,1,-1\rangle$ of R^3. Find $[\mathbf{v}]_{\mathrm{B}}$ for $\mathbf{v}=\langle 2,-7,3\rangle$.

A $\langle 0,-2,-3\rangle$

B $\langle -8,-4,1\rangle$

C $\langle 0,2,3\rangle$

D $\langle 2,-7,3\rangle$